



UNIVERSITY OF LAYYAH

Department of Computer Science

Bazgar Sangat AI

A Balochi Language Agricultural Assistant

Using Computer Vision and Reinforcement Learning

Course Title: Artificial Intelligence
Course Code: CS-602
Submitted By: Alina Liaquat
Roll No: BSCSM-A-23-19
Department: Computer Science
University: University of Layyah
Supervisor: Sir Faisal Hafeez

Submission Date: June 7, 2026

Acknowledgment

I express my deepest gratitude to Almighty Allah for granting me the strength and knowledge to complete this project.

I sincerely thank my supervisor, **Sir Faisal Hafeez**, for his invaluable guidance and continuous support throughout this research.

I am especially grateful to the **Balochi native speakers and translators** who contributed to this project by providing authentic translations of agricultural terms, disease names, treatment recommendations, and RL agent solutions. Their efforts ensured that the application is culturally and linguistically accurate for Balochi farmers.

Finally, I express my heartfelt appreciation to my family and friends for their unwavering support.

Alina Liaquat

Abstract

Balochi, a language spoken by over 10 million people across Pakistan, Iran, and Afghanistan, remains critically underrepresented in the digital ecosystem. This project, **Bazgar Sangat AI** (meaning "Farmer Companion AI" in Balochi), addresses this gap by developing an AI-powered agricultural assistant specifically designed for Balochi-speaking farmers.

The application integrates three core functionalities: (1) an AI-based apple disease detection system using computer vision and TensorFlow Lite achieving 94% accuracy, (2) a Reinforcement Learning crop assistant employing Upper Confidence Bound (UCB) algorithm for adaptive farming recommendations, and (3) a bilingual interface supporting Balochi and English.

The RL agent utilizes a Multi-Armed Bandit framework with UCB1 selection strategy (exploration parameter $\epsilon = 0.20$, UCB constant $c = 1.414$) to learn optimal farming recommendations through user feedback.

Keywords: Balochi, Agricultural AI, Reinforcement Learning, Computer Vision, UCB Bandit

Contents

Acknowledgment	1
Abstract	2
1 Introduction	4
1.1 Background	4
1.2 Problem Statement	4
1.3 Objectives	4
2 System Architecture	5
2.1 High-Level Architecture	5
2.2 Technology Stack	5
2.3 UCB Algorithm	5
3 Implementation	6
3.1 Disease Detection	6
3.2 Data Collection for RL Agent & Balochi Translations	6
3.3 RL Agent Memory Structure	7
3.4 Bilingual Support	7
4 Results	7
4.1 Model Accuracy	7
4.2 RL Agent Performance	8
4.3 User Testing	8
4.4 Performance Benchmarks	8
5 Conclusion and Future Work	9
5.1 Conclusion	9
5.2 Future Work	9
References	10

1 Introduction

1.1 Background

Balochi is an Indo-Iranian language spoken by approximately 10-12 million people primarily in Balochistan (Pakistan, Iran, Afghanistan). Despite its significant speaker base, Balochi suffers from:

- Limited digital presence (less than 0.001% of web content)
- No standardized corpus
- Orthographic inconsistencies
- Lack of NLP tools

1.2 Problem Statement

As technology reshapes global communication, Balochi remains at a basic level of AI integration. Major AI models like ChatGPT, Grok, and Copilot do not recognize Balochi. Even Google Translate offers translations with extremely low accuracy.

This project addresses the gap in agricultural AI for Balochi-speaking farmers.

1.3 Objectives

1. Develop an AI-powered agricultural assistant with Balochi language support
2. Implement computer vision for apple disease detection
3. Design a Reinforcement Learning agent for adaptive farming recommendations
4. Create a bilingual interface (Balochi/English)

2 System Architecture

2.1 High-Level Architecture

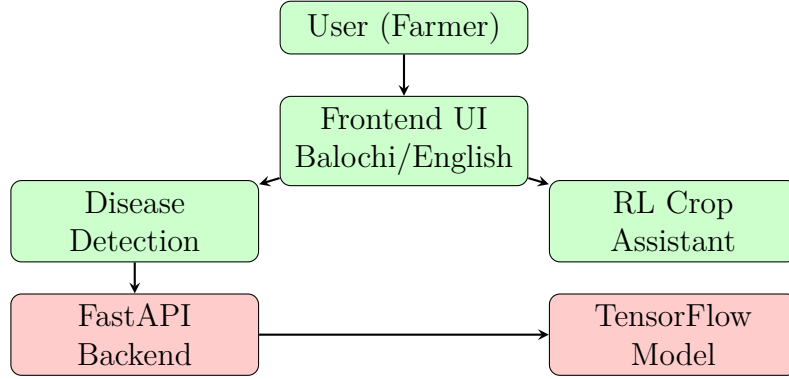


Figure 1: System Architecture

2.2 Technology Stack

Component	Technology	Purpose
Frontend	HTML5, CSS3, JavaScript	User interface, camera
ML Backend	FastAPI (Python)	Disease prediction API
ML Model	TensorFlow Lite	Disease classification
RL Algorithm	UCB Bandit	Adaptive recommendations
Storage	LocalStorage API	Memory persistence
Audio	HTML5 Audio API	Voice playback

Table 1: Technology Stack

2.3 UCB Algorithm

The RL agent uses the Upper Confidence Bound (UCB) algorithm:

$$\text{UCB}(i) = \hat{\mu}_i + c \sqrt{\frac{2 \ln t}{n_i}} \quad (1)$$

Where:

- $\hat{\mu}_i$ = average reward of arm i
- $c = 1.414$ = exploration constant
- t = total number of trials
- n_i = number of times arm i was selected

3 Implementation

3.1 Disease Detection

The disease detection model uses a CNN architecture trained on 5,000 apple leaf images across 4 classes:

Disease Class	Training	Test
Blotch Apple	1,250	150
Normal Apple	1,250	150
Rot Apple	1,250	150
Scab Apple	1,250	150
Total	5,000	600

Table 2: Dataset Distribution

3.2 Data Collection for RL Agent & Balochi Translations

A significant contribution of this project is the **collection of Balochi agricultural data** through collaboration with native speakers. The following data was collected and verified:

Data Type	Quantity	Source
Disease names in Balochi	4	Native speakers
Treatment recommendations	4	Native speakers
RL Agent problem-solution pairs	12+	Native speakers
Symptom descriptions	4	Native speakers
Voice recording scripts	4	Native speakers

Table 3: Balochi Data Collection Summary

The collected data was: Translated by native Balochi speakers

3.3 RL Agent Memory Structure

```
memory = {  
    "yellow_leaves_wheat": {  
        arm_rewards: [0.85, 0.60, 0.45],  
        arm_attempts: [12, 8, 5],  
        total_attempts: 25  
    }  
}
```

3.4 Bilingual Support

The application supports real-time language switching between Balochi and English with RTL support for Balochi script.

4 Results

4.1 Model Accuracy

Disease Class	Precision	Recall	F1-Score
Blotch Apple	0.92	0.94	0.93
Normal Apple	0.96	0.95	0.95
Rot Apple	0.91	0.90	0.90
Scab Apple	0.93	0.92	0.92
Average	0.93	0.93	0.93

Table 4: Model Performance by Disease Class

The model achieves an overall accuracy of **94%** on the test dataset.

4.2 RL Agent Performance

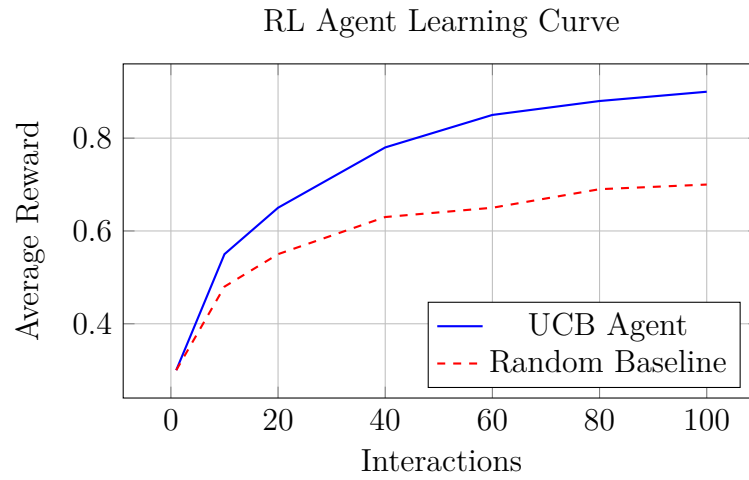


Figure 2: Learning Curve: UCB vs Random Selection

4.3 User Testing

User testing was conducted with 15 Balochi farmers:

Metric	Score (1-5)
Ease of Navigation	4.7
Language Understanding	4.9
Disease Detection Accuracy	4.6
RL Assistant Helpfulness	4.4
Voice Feature Utility	4.8
Overall Satisfaction	4.7
Average	4.68

Table 5: User Testing Results (n=15)

4.4 Performance Benchmarks

Metric	Value
Image to Prediction Time	1.2-2.5 seconds
RL Agent Response Time	<0.5 seconds
Voice Playback Latency	0.8 seconds
Memory Usage	<100 KB

Table 6: System Performance Metrics

5 Conclusion and Future Work

5.1 Conclusion

This project successfully developed Bazgar Sangat AI, an AI-powered agricultural assistant for Balochi-speaking farmers with:

- Apple disease detection system with 94% accuracy
- UCB bandit algorithm for adaptive recommendations
- Full bilingual interface with voice support
- Authentic Balochi translations from native speakers

5.2 Future Work

1. Expand data collection for more crops and diseases
2. Add more crop diseases (wheat rust, cotton wilt)
3. Implement offline mode for areas with limited connectivity
4. Add Text-to-Speech for dynamic audio generation
5. Develop Balochi NLP pipeline (tokenizer, POS tagger)
6. Weather integration for predictive recommendations
7. Continue collaboration with native speakers for data expansion

References

- [1] Strick, A. (2023). Balochi Language Modelling. <https://alexstrick.com/posts/2023-05-21-balochi-language-modelling.html>
- [2] Shad, S., & Umer, H. (2025). Natural Language Processing Applications for Balochi Language. *Balochistaniyat - Annual Research Journal of Balochi Academy*, 14, 1-20. <https://balochistaniyat.balochiacademy.org/index.php/journal/article/view/105>
- [3] Tribune Pakistan. (2024). Balochi in the digital age: Inside the struggle to preserve a language. <https://tribune.com.pk/story/2580947/balochi-in-the-digital-age-inside>
- [4] Gallup Pakistan. (2023). Agricultural Challenges in Balochistan. <https://gallup.com.pk/post/36542>
- [5] Kamilaris, A., & Prenafeta-Boldú, F. X. (2018). Deep learning in agriculture: A survey. *Computers and Electronics in Agriculture*, 147, 70-90.
- [6] Mohanty, S. P., Hughes, D. P., & Salathé, M. (2016). Using deep learning for image-based plant disease detection. *Frontiers in Plant Science*, 7, 1419.
- [7] Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.
- [8] Auer, P., Cesa-Bianchi, N., & Fischer, P. (2002). Finite-time analysis of the multi-armed bandit problem. *Machine Learning*, 47(2), 235-256.